

SAN ANTONIO STINSON, TX, USA

WMO: 722523

Lat: 29.339N

Lon: 98.472W

Elev: 571

StdP: 14.40

Time Zone: -6.00 (NAC)

Period: 98-19

WBAN: 12970

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	31.0	34.0	11.8	10.3	47.0	16.0	12.6	49.4	21.3	56.1	19.3	57.6	5.1	10	0.415

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.4	101.3	73.9	99.4	74.2	97.5	73.9	78.7	89.2	78.0	88.5	77.3	87.8	9.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.3	140.4	81.6	75.4	135.9	80.9	74.9	133.6	80.7	42.7	89.9	41.7	89.2	41.1	87.6	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	17.2	15.8	24.6	103.8	3.1	3.1	22.4	106.1	20.6	107.9	18.9	109.7	16.6	111.9
			22.2	79.6	2.8	1.1	20.2	80.4	18.6	81.1	17.0	81.7	15.0	82.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	71.1	53.6	58.2	64.6	71.4	78.1	83.9	85.4	86.4	81.4	72.7	62.5	54.7	(6)
		DBStd	13.55	8.79	9.86	8.13	6.67	5.44	3.60	3.37	3.20	4.96	7.72	8.91	8.72	(7)
		HDD50	163	63	33	6	1	0	0	0	0	1	11	48		(8)
		HDD65	1238	364	229	109	21	2	0	0	0	26	149	338		(9)
		CDD50	7882	176	263	460	644	870	1016	1096	1129	943	703	388	194	(10)
		CDD65	3480	11	40	97	214	406	566	631	664	493	264	75	19	(11)
		CDH74	37734	94	270	644	1704	3887	6582	7727	8535	5324	2356	506	105	(12)
		CDH80	18328	9	55	153	577	1664	3359	4174	4794	2577	860	97	9	(13)
(14)	Wind	WSAvg	7.0	6.5	7.3	7.9	8.3	8.6	8.0	7.2	6.5	5.8	6.1	6.0	6.1	(14)
(15)	Precipitation	PrecAvg	31.1	1.9	1.4	2.0	2.6	4.1	3.2	2.7	2.2	3.2	3.4	2.2	1.7	(15)
(16)		PrecMax	48.1	5.4	4.3	7.1	6.7	9.4	11.3	11.7	8.6	9.5	12.2	9.2	6.2	(16)
(17)		PrecMin	12.6	0.1	0.2	0.0	0.2	0.6	0.4	0.0	0.0	0.0	0.0	0.0	0.3	(17)
(18)		PrecStd	11.0	1.4	1.3	1.5	1.9	2.5	2.5	3.2	2.8	2.6	3.5	2.6	1.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	81.0	86.2	88.3	94.9	98.1	102.3	103.0	103.5	101.6	93.7	86.8	80.9		(19)
(20)		MCWB	61.7	62.7	66.2	70.4	74.2	74.5	73.9	73.3	72.3	73.0	70.5	66.1		(20)
(21)		DB	76.6	81.0	84.0	89.7	94.3	99.2	100.5	101.3	97.9	90.6	82.5	77.1		(21)
(22)		MCWB	60.8	63.9	65.7	68.4	73.2	75.1	74.3	73.6	72.8	72.5	67.5	63.9		(22)
(23)		DB	72.7	77.2	80.9	86.1	91.4	96.6	98.8	99.5	94.8	88.0	79.8	73.3		(23)
(24)		MCWB	60.3	62.4	66.0	68.6	73.4	74.4	74.3	74.0	73.4	71.1	66.5	63.5		(24)
(25)		DB	69.5	73.1	77.4	82.9	88.8	93.5	96.4	97.5	92.0	85.2	76.5	70.1		(25)
(26)		MCWB	59.8	62.1	63.9	68.1	72.9	74.3	74.4	74.0	73.4	70.1	66.1	61.6		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	68.7	70.3	71.5	75.9	78.4	80.1	79.8	79.5	79.0	78.2	73.8	70.5		(27)
(28)		MCDB	72.9	77.8	80.4	86.1	89.5	91.8	90.6	87.8	87.8	85.4	79.8	76.1		(28)
(29)		WB	66.9	68.5	70.0	73.8	77.1	78.6	78.3	78.2	77.7	76.5	72.1	68.8		(29)
(30)		MCDB	71.7	74.7	77.4	83.1	88.0	89.9	89.3	88.4	86.8	83.5	78.0	73.5		(30)
(31)		WB	64.3	67.1	69.0	72.4	75.9	77.7	77.5	77.4	77.0	75.3	70.4	66.9		(31)
(32)		MCDB	69.4	72.7	75.6	80.8	85.7	88.4	88.3	88.6	85.9	82.2	76.3	71.1		(32)
(33)		WB	61.6	65.4	67.8	71.3	75.0	76.9	76.9	76.7	76.2	74.0	68.6	63.8		(33)
(34)		MCDB	67.5	70.7	74.2	78.9	84.1	87.0	87.3	87.5	84.9	80.9	74.5	68.3		(34)
(35)	Mean Daily Temperature Range	MDBR	21.9	21.3	20.9	20.7	19.6	19.6	20.4	21.4	19.8	21.0	21.1	21.4		(35)
(36)		5% DB	27.7	25.2	23.4	24.4	22.4	23.1	24.3	24.2	23.2	21.4	22.5	26.0		(36)
(37)		MCWBR	15.6	13.3	10.9	10.1	6.9	4.9	4.8	4.8	5.6	7.8	11.2	15.1		(37)
(38)		5% WB	20.4	18.2	18.3	17.6	18.3	18.4	19.5	20.2	18.1	17.0	17.4	19.1		(38)
(39)		MCWBR	14.3	12.0	10.3	8.9	7.1	5.6	4.8	4.7	5.5	8.0	10.6	14.0		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.331	0.337	0.361	0.404	0.451	0.432	0.439	0.424	0.417	0.375	0.363	0.321		(40)
(41)		taud	2.488	2.476	2.421	2.317	2.233	2.340	2.353	2.378	2.350	2.453	2.423	2.529		(41)
(42)		Ebn at Noon	288	296	294	283	269	272	270	274	272	279	273	286		(42)
(43)		Edh at Noon	29	32	36	41	45	40	40	38	38	32	31	27		(43)
(44)	All-Sky Solar Radiation	RadAvg	958	1125	1423	1689	1831	2094	2108	2018	1656	1375	1033	851		(44)
(45)		RadStd	108	203	148	170	187	133	184	115	166	180	115	110		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	+0.32	N/A	N/A	N/A	+457	+338
(47)	Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+193	+162

Nomenclature: See separate page